

Dokumentasi Proyek DurianTrace Backend

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1. Ringkasan Proyek

DurianTrace Backend adalah API Laravel 12 untuk aplikasi pelacakan durian. Backend ini menangani registrasi dan login berbasis role, profil per role, data kebun dan batch panen, verifikasi dan shipment oleh pengepul, receipt distributor, katalog dan order UMKM, transaksi konsumen, serta public trace tanpa login.

2. Standar Umum API

Komponen	Keterangan
Format	REST API JSON
Authentication	Laravel Sanctum Bearer Token
Content-Type	application/json untuk request JSON
Accept	application/json
Envelope response	success, message, data
Validation error	success = false dan errors berisi detail field

Contoh response sukses

```
{
  "success": true,
  "message": "OK",
  "data": {}
}
```

Contoh response error validasi

```
{
  "success": false,
  "message": "Data tidak valid",
  "errors": {
    "email": ["Email wajib diisi."]
  }
}
```

3. Role Pengguna

Role	Deskripsi
petani	Mengelola kebun, batch panen, dan data profil petani
pengepul	Melihat stok, verifikasi batch, membuat shipment, dan memproses shipment
distributor	Menerima shipment dan membuat receipt
umkm	Mengelola produk dan order UMKM
konsumen	Melihat produk konsumen dan membuat transaksi

4. Aturan Penting

Aturan	Penjelasan
Role tunggal	Satu user hanya memiliki satu role aktif
Kode stabil	Kode domain tidak boleh berubah setelah data dibuat
Traceability	Setiap perubahan penting dicatat sebagai event batch
Public trace	Endpoint trace dapat dibuka tanpa login
Payload konsisten	Frontend selalu menerima envelope JSON yang seragam

5. Endpoint Auth

Registrasi pengguna

Metode	POST
Path	/api/register
Auth	Public
Header	Accept: application/json Content-Type: application/json
Request Body	{ "first_name": "Risqi", "last_name": "Firdaus", "phone": "81234567890", "email": "risqi@example.com", "password": "secret123", "password_confirmation": "secret123", "role": "petani" }
Response Sukses	{ "success": true, "message": "Registrasi berhasil", "data": { "user": { "...": "..."}, "token": "1 xxxxxxxxxxxxxxxx", "dashboard": "petani" } }

Catatan Role harus salah satu dari petani, pengepul, distributor, umkm, konsumen.
Status sukses: 201.

Login pengguna

Metode POST

Path /api/login

Auth Public

Header Accept: application/json Content-Type: application/json

Request Body { "identifier": "risqi@example.com", "password": "secret123", "role": "petani" }

Response Sukses { "success": true, "message": "Login berhasil", "data": { "user": { "...": "...", "token": "1|xxxxxxxxxxxxxxxx", "dashboard": "petani" } }

Catatan Identifier dapat berupa email, username, atau phone sesuai policy backend.
Error 401/403/404/422 memakai message yang mudah dibaca.

Session aktif

Metode GET

Path /api/me

Auth Bearer token

Header Accept: application/json Authorization: Bearer <token>

Request Body Tidak ada body request.

Response Sukses { "success": true, "message": "Session aktif", "data": { "user": { "...": "...", "token": "1|xxxxxxxxxxxxxxxx", "dashboard": "petani" } }

Catatan Dipakai untuk restore session. Token tidak valid akan menghasilkan 401.

Logout

Metode POST

Path /api/logout

Auth Bearer token

Header Accept: application/json Authorization: Bearer <token>

Request Body Tidak ada body request.

Response Sukses { "success": true, "message": "Logout berhasil", "data": null }

Catatan Token aktif dihapus dari server jika memungkinkan.

6. Endpoint Profil per Role

Role	GET	PUT
petani	/api/farmer/profile	/api/farmer/profile
pengumpul	/api/collector/profile	/api/collector/profile
distributor	/api/distributor/profile	/api/distributor/profile
umkm	/api/umkm/profile	/api/umkm/profile
konsumen	/api/consumer/profile	/api/consumer/profile

Request body profil umum

```
{  
  "full_name": "Risqi Firdaus Setiawan",  
  "contact": "081234567890",  
  "email": "risqi@example.com",  
  "location": "Desa Pakis, Kabupaten Jember"  
}
```

Catatan: backend bisa menerima field profil sesuai role, misalnya business_name, owner_name, display_name, address, contact, avatar_path.

7. Endpoint Petani

Daftar kebun

Metode	GET
Path	/api/farmer/farms
Auth	Bearer token
Header	Accept: application/json Authorization: Bearer <token>

Request Tidak ada body request.

Body

Response { "success": true, "message": "OK", "data": [{ "id": 1, "farmer_id":
Sukses 10, "name": "Kebun Pakis 1", "province": "Jawa Timur", "city":
"Kabupaten Jember", "district": "Pakis", "village": "Pakis", "address":
"Jl. Raya Pakis No. 1", "latitude": null, "longitude": null }] }

Catatan Hanya data kebun milik petani yang login.

Tambah kebun

Metode POST

Path /api/farmer/farms

Auth Bearer token

Header Accept: application/json Content-Type: application/json Authorization: Bearer
<token>

Request { "name": "Kebun Pakis 2", "province": "Jawa Timur", "city": "Kabupaten
Body Jember", "district": "Pakis", "village": "Pakis", "address": "Jl. Raya Pakis No.
2", "latitude": -8.1234, "longitude": 113.5678 }

Response { "success": true, "message": "Kebun berhasil dibuat", "data": { "...": "..." } }
Sukses

Catatan Latitude dan longitude opsional tetapi harus numerik jika diisi.

Daftar batch

Metode GET

Path /api/farmer/batches

Auth Bearer token

Header Accept: application/json Authorization: Bearer <token>

Request Tidak ada body request.

Body

Response { "success": true, "message": "OK", "data": [{ "code": "DRN-2026-
Sukses 000128", "farmer_id": 10, "farm_id": 2, "farm_name_snapshot":
"Kebun Pakis 2", "variety": "Montong", "grade": "A", "quantity_kg": 52,
"unit": "kg", "fruit_count": 18, "harvest_date": "2026-07-20", "status":

```
"created"  }  ] }
```

Catatan List diurutkan dari data terbaru.

Tambah batch

Metode POST

Path /api/farmer/batches

Auth Bearer token

Header Accept: application/json Content-Type: application/json Authorization: Bearer <token>

Request { "farm_id": 2, "variety": "Montong", "grade": "A", "quantity_kg": 52,
Body "unit": "kg", "fruit_count": 18, "harvest_date": "2026-07-20", "fertilizer":
"Organik Kompos", "harvest_method": "Jatuh Alami", "maturity_level":
"Matang Pohon", "shelf_life_estimate": "2-3 hari", "storage_suggestion":
"Simpan di tempat sejuk.", "notes": "Kulit utuh, aroma kuat.", "photo_path":
"uploads/batches/batch-001.jpg" }

Response { "success": true, "message": "Batch berhasil dibuat", "data": { "code":
Sukses "DRN-2026-000128", "status": "created", "harvest_date": "2026-07-20" } }

Catatan Backend membuat kode unik stabil dan menulis event Batch Dibuat.

Detail batch

Metode GET

Path /api/farmer/batches/{code}

Auth Bearer token

Header Accept: application/json Authorization: Bearer <token>

Request Body Tidak ada body request.

Response { "success": true, "message": "OK", "data": { "batch": { "...": "..."},
Sukses "events": [{ "...": "..." }] } }

Catatan Dipakai oleh detail batch dan timeline trace.

Update batch

Metode	PATCH
Path	/api/farmer/batches/{code}
Auth	Bearer token
Header	Accept: application/json Content-Type: application/json Authorization: Bearer <token>
Request Body	{ "farm_id": 2, "variety": "Montong", "grade": "A", "quantity_kg": 55, "fruit_count": 19, "harvest_date": "2026-07-20", "notes": "Updated after correction window." }
Response Sukses	{ "success": true, "message": "Batch berhasil diperbarui", "data": { "code": "DRN-2026-000128", "status": "created" } }
Catatan	Hanya boleh saat status masih created.

8. Endpoint Pengepul

Ringkasan stok

Metode	GET
Path	/api/collector/stock
Auth	Bearer token
Header	Accept: application/json Authorization: Bearer <token>
Request Body	Tidak ada body request.
Response Sukses	{ "success": true, "message": "OK", "data": { "active_batch_count": 2, "total_weight_kg": 158, "total_fruit_count": 38, "grade_breakdown": [{ "...": "..." }], "variety_breakdown": [{ "...": "..." }] } }
Catatan	Menampilkan ringkasan stok aktif milik pengepul.

Daftar shipment

Metode	GET
Path	/api/collector/shipment-batches
Auth	Bearer token

Header Accept: application/json Authorization: Bearer <token>

Request Tidak ada body request.

Body

Response { "success": true, "message": "OK", "data": [{ "code": "PGL-2026-
Sukses 000001", "collector_id": 20, "source_batch_codes": ["DRN-2026-000128"],
"total_weight_kg": 158, "total_fruit_count": 38, "status": "readyToShip",
"destination_type": "distributor", "packaged_at": "2026-07-20T10:00:00Z" }
] }

Catatan List diurutkan dari data terbaru.

Tambah shipment

Metode POST

Path /api/collector/shipment-batches

Auth Bearer token

Header Accept: application/json Content-Type: application/json Authorization: Bearer
<token>

Request { "source_batch_codes": ["DRN-2026-000128", "DRN-2026-000110"],
Body "destination_type": "distributor", "warehouse_note": "Pengiriman via truk
pendingin." }

Response { "success": true, "message": "Shipment berhasil dibuat", "data": { "code":
Sukses "PGL-2026-000001", "status": "readyToShip" } }

Catatan Backend memvalidasi batch, menyimpan snapshot, dan membuat kode
shipment stabil.

Kirim shipment

Metode PATCH

Path /api/collector/shipment-batches/{code}/send

Auth Bearer token

Header Accept: application/json Content-Type: application/json Authorization: Bearer
<token>

Request Tidak ada body request.

Body

Response { "success": true, "message": "Shipment terkirim", "data": { "code": "PGL-Sukses 2026-000001", "status": "sent", "sent_at": "2026-07-20T11:00:00Z" } }

Catatan Hanya boleh jika status shipment readyToShip.

Selesaikan shipment

Metode PATCH

Path /api/collector/shipment-batches/{code}/complete

Auth Bearer token

Header Accept: application/json Content-Type: application/json Authorization: Bearer <token>

Request Tidak ada body request.

Body

Response { "success": true, "message": "Shipment selesai", "data": { "code": "PGL-Sukses 2026-000001", "status": "completed", "completed_at": "2026-07-20T13:00:00Z" } }

Catatan Hanya boleh jika status shipment sent.

Verifikasi batch

Metode POST

Path /api/collector/batches/{code}/verify

Auth Bearer token

Header Accept: application/json Content-Type: application/json Authorization: Bearer <token>

Request { "received_quantity_kg": 94, "received_fruit_count": 17,
Body "grade_breakdown": [{ "grade": "A", "weight_kg": 60, "fruit_count": 11 }, { "grade": "B", "weight_kg": 34, "fruit_count": 6 }], "quality_notes": "Mayoritas grade A.", "verified_by": "Pengepul Jember" }

Response { "success": true, "message": "Batch terverifikasi", "data": { "code": "DRN-Sukses 2026-000128", "status": "verifiedByCollector" } }

Catatan Menyimpan grade breakdown dan event verifikasi.

Tolak batch

Metode	POST
Path	/api/collector/batches/{code}/reject
Auth	Bearer token
Header	Accept: application/json Content-Type: application/json Authorization: Bearer <token>
Request Body	{ "reason": "Beberapa buah retak dan tingkat kematangan tidak seragam.", "rejected_by": "Pengepul Jember" }
Response Sukses	{ "success": true, "message": "Batch ditolak", "data": { "code": "DRN-2026-000128", "status": "rejected" } }
Catatan	Batch yang ditolak berhenti dari alur mutasi berikutnya.

9. Endpoint Distributor

Daftar shipment distributor

Metode	GET
Path	/api/distributor/shipments
Auth	Bearer token
Header	Accept: application/json Authorization: Bearer <token>
Request Body	Tidak ada body request.
Response Sukses	{ "success": true, "message": "OK", "data": [{ "code": "PGL-2026-000001", "collector_id": 20, "destination_type": "distributor", "status": "sent", "total_weight_kg": 158, "total_fruit_count": 38 }] }
Catatan	Hanya shipment dengan destination_type = distributor.

Detail shipment distributor

Metode	GET
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Path	/api/distributor/shipments/{code}
Auth	Bearer token
Header	Accept: application/json Authorization: Bearer <token>
Request	Tidak ada body request.
Body	
Response	{ "success": true, "message": "OK", "data": { "shipment": { "...": "..." },
Sukses	"source_batches": [{ "...": "..." }], "receipt": null } }
Catatan	Frontend dapat menampilkan shipment, source batch, dan receipt dalam satu layar detail.

Simpan receipt

Metode	POST
Path	/api/distributor/shipments/{code}/receipt
Auth	Bearer token
Header	Accept: application/json Content-Type: application/json Authorization: Bearer <token>
Request	{ "received_weight_kg": 156, "received_fruit_count": 37, "condition":
Body	"minorDamage", "discrepancy_note": "Selisih kecil karena sortasi ulang.",
	"quality_note": "Barang masih layak jual." }
Response	{ "success": true, "message": "Receipt berhasil disimpan", "data": {
Sukses	"shipment_code": "PGL-2026-000001", "condition": "minorDamage",
	"received_at": "2026-07-20T13:30:00Z" } }
Catatan	Jika jumlah diterima berbeda dari jumlah diharapkan, discrepancy_note wajib diisi.

10. Endpoint UMKM

Daftar produk

Metode	GET
Path	/api/umkm/products

Auth Bearer token

Header Accept: application/json Authorization: Bearer <token>

Request Tidak ada body request.

Body

Response { "success": true, "message": "OK", "data": [{ "code": "UMKM-P-001",
Sukses "name": "Pancake Durian Premium", "category": "Olahan", "status":
"aktif", "price_label": "Rp 68.000", "stock_label": "Stok 24 paket",
"description": "Pancake durian lembut.", "qr_code_data": "UMKM-P-001" }
] }

Catatan Produk adalah milik UMKM yang sedang login.

Tambah produk

Metode POST

Path /api/umkm/products

Auth Bearer token

Header Accept: application/json Content-Type: application/json Authorization: Bearer <token>

Request { "name": "Pancake Durian Premium", "category": "Olahan", "price_label":
Body "Rp 68.000", "stock_label": "Stok 24 paket", "description": "Pancake durian
lembut.", "status": "aktif", "qr_code_data": "UMKM-P-001", "image_path":
"uploads/products/p-001.jpg" }

Response { "success": true, "message": "Produk berhasil dibuat", "data": { "code":
Sukses "UMKM-P-001", "status": "aktif" } }

Catatan Backend menerima price_label dan stock_label lalu mengonversinya ke data numerik.

Daftar order

Metode GET

Path /api/umkm/orders

Auth Bearer token

Header Accept: application/json Authorization: Bearer <token>

Request Tidak ada body request.

Body

Response { "success": true, "message": "OK", "data": [{ "id": "ORD-2026-0001",
Sukses "product_name": "Pancake Durian Premium", "buyer_name": "Rina Saputri",
"quantity": 2, "total_label": "Rp 136.000", "status": "diproses",
"created_at": "2026-07-20T10:00:00Z", "qr_code_data": "ORD-2026-0001"
}] }

Catatan Urutan data dari yang terbaru.

Tambah order

Metode POST

Path /api/umkm/orders

Auth Bearer token

Header Accept: application/json Content-Type: application/json Authorization: Bearer
<token>

Request { "product_id": 1, "buyer_name": "Rina Saputri", "quantity": 2,
Body "total_label": "Rp 136.000", "status": "diproses", "qr_code_data": "ORD-
2026-0001", "note": "Bayar di tempat." }

Response { "success": true, "message": "Order berhasil dibuat", "data": { "id": "ORD-
Sukses 2026-0001", "status": "diproses" } }

Catatan Backend menerima product_id dan total_label, lalu menyimpan order dengan
kode stabil.

11. Endpoint Konsumen

Daftar produk konsumen

Metode GET

Path /api/consumer/products

Auth Bearer token

Header Accept: application/json Authorization: Bearer <token>

Request Tidak ada body request.

Body

Response { "success": true, "message": "OK", "data": [{ "code": "UMKM-P-001",
Sukses "name": "Pancake Durian Premium", "category": "Paket", "status":
"readyToSell", "price_label": "Rp 68.000", "short_description": "Paket isi 4
potong.", "umkm_name": "UMKM Sari Durian Jember", "location":
"Kabupaten Jember, Jawa Timur", "rating": 4.9, "stock_label": "Stok 24
paket" }] }

Catatan Hanya produk yang terlihat untuk konsumen.

Detail produk konsumen

Metode GET

Path /api/consumer/products/{code}

Auth Bearer token

Header Accept: application/json Authorization: Bearer <token>

Request Tidak ada body request.

Body

Response { "success": true, "message": "OK", "data": { "product": { "code":
Sukses "UMKM-P-001", "name": "Pancake Durian Premium",
"source_batch_code": "DRN-2026-000119", "source_variety": "Montong",
"source_grade": "B" } } }

Catatan Dipakai oleh layar detail konsumen untuk menampilkan trace sumber produk.

Tambah transaksi konsumen

Metode POST

Path /api/consumer/transactions

Auth Bearer token

Header Accept: application/json Content-Type: application/json Authorization: Bearer
<token>

Request { "product_id": 1, "quantity": 2, "buyer_address": "Desa Pakis, Kec. Panti, Kab.
Body Jember", "buyer_coordinates": "-8.2285, 113.6204", "payment_method":
"QRIS", "payment_status": "unpaid", "bank_name": null, "account_number":
null, "note": "Menunggu pembayaran." }

Response { "success": true, "message": "Transaksi berhasil dibuat", "data": { "id":
Sukses "TRX-2026-0001", "status": "processing", "payment_status": "unpaid",
"qr_code_data": "TRX-2026-0001" } }

Catatan Jika product_id kosong, backend akan menolak request dengan validasi.

Daftar transaksi konsumen

Metode GET

Path /api/consumer/transactions

Auth Bearer token

Header Accept: application/json Authorization: Bearer <token>

Request Tidak ada body request.

Body

Response { "success": true, "message": "OK", "data": [{ "id": "TRX-2026-0001",
Sukses "status": "processing", "payment_status": "unpaid", "created_at": "2026-
07-20T10:30:00Z" }] }

Catatan Urutan data dari yang terbaru.

12. Public Trace API

Trace batch publik

Metode GET

Path /api/trace/{batchCode}

Auth Public

Header Accept: application/json

Request Tidak ada body request.

Body

Response { "success": true, "message": "OK", "data": { "batch": { "code": "DRN-
Sukses 2026-000128", "variety": "Montong", "grade": "A", "status":
"verifiedByCollector", "farm_name_snapshot": "Kebun Pakis 2",
"farm_location": "Kabupaten Jember, Jawa Timur" }, "events": [{ "...": "..." }
], "source_batches": [], "shipment_history": [], "public_url":

```
"https://domain-produk/api/trace/DRN-2026-000128" }}
```

Catatan Endpoint ini penting untuk QR scan tanpa login. Response berisi timeline dan URL publik yang stabil.

13. Catatan Maintenance

Backend sekarang memakai helper response dan formatter terpusat. Jika nanti kontrak frontend berubah, biasanya cukup menyesuaikan di satu tempat: helper response, formatter payload, atau controller yang bersangkutan.